

A Multi-Agent Resource Allocation Method for Unmanned Aerial Vehicle Networks based on Adaptive Spatial Reward Mechanisms and Context-Awareness

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Abstract

In the unmanned aerial vehicle (UAV) communication network, dynamic resource management is crucial for the efficient allocation and management of network resources. However, the limitations of transmission power and battery capacity faced by UAVs, as well as the complexity and variability of the communication environment, pose significant challenges to resource management. Particularly in UAV communication, the communication resource allocation and management has become an urgent issue. Therefore, the article proposes a method for resource allocation in multi-UAV networks, namely the Multi-agent Reinforcement Learning based on Adaptive Space Reward Mechanism and Agent Contextual Awareness (ASAC). Firstly, the UAV network is modeled based on the Multi-Agent Reinforcement Learning (MARL) framework for multi-UAV networks, which includes communication entities, channels, user locations, and UAV power, among other elements. Secondly, the adaptive space reward mechanism and agent contextual awareness technology is used to enhance the adaptability and stability of the model through real-time adjustment of the distance influence factor between agents, moreover, the efficiency of model training is significantly improved by utilizing the proximity information between agents. Finally, relevant experiments are conducted to analyze and discuss the impact of different parameters on network communication efficiency. Compared with the baseline methods, the results

show that the article outperforms existing baseline algorithms in terms of resource allocation efficiency, especially in handling high number users networks, where the method of ASAC improves resource allocation efficiency by 16%. Therefore, the method is expected to be applied in the field of wireless communication, providing new breakthroughs for the development of related technologies.

CCS Concepts

• Computing methodologies → Multi-agent systems.

Keywords

Multi-agent reinforcement learning(MARL), unmanned aerial vehicle networks(UAV network), unmanned aerial vehicle communications(UAV communications), resource allocation

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1 Introduction

The innovation and development of aerial communication networks have spurred the widespread deployment of wireless infrastructure. In recent years, they have garnered significant attention for their ability to deliver higher network capacity and broader coverage [3]. Unmanned Aerial Vehicles, also referred to as remote-controlled aircraft systems or UAVs, are small unmanned aerial vehicles that utilize Third-Generation Partnership Project (3GPP) technology for rapid deployment to enhance ground network coverage and

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were increasingly valued by researchers[12]. Compared to traditional ground communication channels, the communication link from UAVs to ground stations tends to exhibit Line-of-Sight (LoS) characteristics[1], which is particularly beneficial for optimizing wireless communication connections.

Specifically, diverse aerial platforms constructed based on UAVs have become a focal point of interest in both academic and industrial circles, with active researches focusing on deployment strategies, navigation technology, and the design and implementation of control systems[17]. In UAV networks, resource allocation issues such as the selection of transmission power, determination of served users, and choice of subchannels are crucial for enhancing the energy efficiency and coverage of UAV communication networks. The energy efficiency and coverage of UAV communication networks, and resource management problems, encompass the optimal configuration of transmission power, allocation of user services, and selection of subchannels, whose decisions directly impact the performance of the network.

1.1 Related work

Unlike traditional terrestrial base stations, UAV deployment is more rapid and configurable. Researchers have explored cooperative UAV deployment strategies to form small-unit networks [24], with Mozaffari and Saad proposing a 3D deployment algorithm based on circular packing to maximize downlink coverage [16]. Carvajal et al. optimized single-UAV 3D placement for user coverage maximization [5], while Hoang et al. developed a continuous placement strategy that minimizes UAV count while ensuring multi-UAV coverage for all ground users [10], demonstrating UAV deployment flexibility and providing references for network optimization.

Beyond deployment, UAV trajectory design for communication performance has gained significant attention. Lu et al. treated UAVs as mobile relays and proposed an SCA-based trajectory optimization method, jointly optimizing power allocation and trajectories to maximize throughput [15]. Fu et al. minimized task completion time in UAV-supported multicast systems by discretizing continuous trajectories [9], while Burhanuddin et al. balanced energy efficiency and latency through joint sub-channel allocation and trajectory design in multi-UAV sensing networks [4]. These studies offer diverse approaches to trajectory optimization for UAV communication.

To reduce human intervention in UAV control, machine learning-based intelligent methods have been explored. Shukla et al. and Zeng et al. developed neural network-driven trajectory design [19, 26], while Kirubakaran et al. proposed a demand-aware UAV deployment method using Gaussian mixture models for predictive accuracy [13]. Ren et al. addressed autonomous path planning with an echo state network-based deep reconstruction algorithm [18], and Chua et al. enhanced multi-UAV communication efficiency via log-linear learning-based joint channel-slot selection [6].

While existing research focuses on UAV deployment and trajectory design [13, 18], resource allocation has emerged as a key topic. Xia et al. and Lu et al. optimized transmission power and subchannel allocation [15, 25], but these studies often ignore time-series correlation and assume full network information access. In practice, UAVs face limited information acquisition and dynamic

environments [21, 22], and centralized methods suffer from scalability issues with network expansion, highlighting critical gaps for future research.

Multi-Agent Reinforcement Learning (MARL) addresses these challenges by enabling distributed intelligent resource management. MARL allows UAVs to optimize strategies via local interactions, reducing computational complexity and enhancing system robustness—key advantages including personalized environmental adaptation, precise local interaction modeling, and distributed processing. Widely applied in cognitive radio and D2D networks [7, 11, 23], MARL offers great potential for UAV communication networks. However, research on MARL-based resource allocation in multi-UAV systems remains scarce, particularly considering UAV mobility and adaptability, indicating a promising direction for future work.

1.2 Motivation and Contribution

The article constructs an efficient MARL framework for multi-UAV networks based on the characteristics of unmanned aerial vehicle (UAV) network resources. The framework by integrating an adaptive spatial reward mechanism and agent context-aware technology aims to efficiently handle the spatiotemporal correlation in multi-UAV systems, through information exchange among agents, it accelerates the system's stability and significantly improves learning efficiency. In the design of the framework, the adaptive spatial reward mechanism allows the algorithm to consider the contribution of neighboring agents with greater weight in the decision-making process, speeding up the convergence rate. The agent context-aware technology, as another core component of the framework, enables real-time sharing of state information and policy choices between adjacent UAVs, promoting global cooperation and effectively enhancing communication efficiency.

2 System model

This section constructs the UAV communication network environment and establishes the resource allocation method for multi-UAV. The UAV communication network environment includes the channel model and the signal model, where the channel model affects the communication mode and path loss between UAV m and ground user l , and the signal model affects the communication quality between UAV m and ground user l . The resource allocation method for multi-UAV includes the actions taken by the UAVs, reward values, etc. The actions determine the resource allocation method of the UAVs, and the reward values measure the efficiency of the total resource allocation. Subsection A defines the UAV-to-ground channel model, subsection B defines the signal model, and subsection C defines the multi-UAV network model, which achieves the resource allocation method for multi-UAV.

There's a multi-UAV downlink communication network as shown in Figure 1, consisting of M UAVs as well as L users, represented by $\mathcal{M} = \{1, 2, 3, \dots, m\}$ and $\mathcal{L} = \{1, 2, \dots, l\}$ respectively. Ground users are allocated within a radius of r_d randomly. The channel model of UAV and ground users is denoted in section A.

When multiple UAVs enter the signal coverage area, they can provide direct communication with ground users. The total bandwidth W that the entire UAV system can operate is divided into $\mathcal{K}$

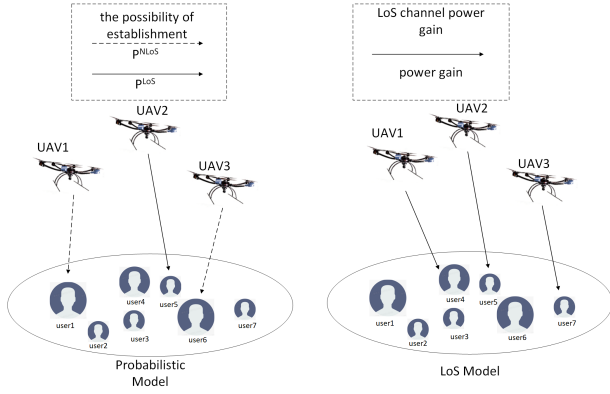


Figure 1: UAV communication networks

orthogonal sub-channels, represented by $K = \{1, 2, \dots, k\}$. The sub-channels occupied by different UAVs may overlap with each other. UAVs fly autonomously according to pre-programmed flight plans, which means the flight trajectories of the UAVs are pre-defined based on the pre-programmed flight plans. As shown in Figure 1, there are three UAVs flying over the considered area according to their pre-defined trajectories. The article explores the dynamic design of resource allocation in multi-UAV networks from the aspects of user selection, power levels, and sub-channel choice. According to the setting of this study, all UAVs communicate without the help of a central controller, and the channel state information (CSI) between UAVs and ground users is known, but UAVs do not have global knowledge of the wireless communication environment. The setting is in line with real-world applications.

2.1 The Channel Model of UAV to Ground

Compared to ground communication, the air-to-ground (A2G) channel of UAVs is highly dependent on factors such as altitude, elevation, and propagation environment [8, 27], with two main models applied in multi-UAV network resource distribution research: the probabilistic model and the LoS model. The probabilistic model considers the different occurrence probabilities of Line-of-Sight (LoS) and Non-Line-of-Sight (NLoS) links, calculating the average path loss by weighting the path losses of the two link types based on their respective probabilities, where path loss includes free space loss and additional environment-related losses, and higher average path loss indicates poorer communication quality. The LoS model provides a reasonable estimate of UAV-to-ground user interactions, with channel gain following the free-space path loss model, which depends on the horizontal positions and altitude of the UAV, the position of the ground user, a reference distance channel power gain, and a path loss exponent.

2.2 Signal Model

In UAV-to-ground transmission, each UAV-ground user pair suffers interference from other UAVs on the same sub-channel. Define $c_m^k(t) = 1$ (UAV m uses sub-channel k) or 0 (otherwise), satisfying:

$$\sum_{k \in \mathcal{K}} c_m^k(t) \leq 1 \quad (1)$$

which means each UAV occupies at most one sub-channel per time slot. Let $a_m^l(t) = 1$ (UAV m serves user l) or 0 (otherwise). The SINR for UAV m communicating with user l on sub-channel k is:

$$\gamma_{m,l}^k(t) = \frac{G_{m,l}^k(t) a_m^l(t) c_m^k(t) P_m(t)}{I_{m,l}^k(t) + \sigma^2} \quad (2)$$

where $G_{m,l}^k(t)$ is the channel gain from UAV m to user l on sub-channel k , $P_m(t)$ is UAV m 's transmission power, $I_{m,l}^k(t)$ is interference:

$$I_{m,l}^k(t) = \sum_{j \in \mathcal{M}, j \neq m} G_{j,l}^k(t) c_j^k(t) P_j(t) \quad (3)$$

and σ^2 is noise. UAV m 's total SINR (used for reward calculation) is:

$$\gamma_m(t) = \sum_{l \in \mathcal{L}} \sum_{k \in \mathcal{K}} \gamma_{m,l}^k(t) \quad (4)$$

UAVs adopt discrete power control: let $\mathbf{P} = \{p_1, \dots, p_J\}$ be power levels, $p_m^j(t) = 1$ (UAV m chooses p_j) or 0 (otherwise), with:

$$\sum_{j \in \mathcal{J}} p_m^j(t) \leq 1, \forall m \in \mathcal{M} \quad (5)$$

Define UAV m 's feasible sets for power, sub-channel, and user selection:

$$\mathcal{P}_m = \left\{ p_m(t) \in \mathbf{P} \mid \sum_{j \in \mathcal{J}} p_m^j(t) \leq 1 \right\}, \forall m \in \mathcal{M} \quad (6)$$

$$\mathcal{C}_m = \{c_m(t) \in \mathcal{K} \mid \sum_{k \in \mathcal{K}} c_m^k(t) \leq 1\}, \forall m \in \mathcal{M} \quad (7)$$

$$\mathcal{A}_m = \{a_m(t) \in \mathcal{L} \mid \sum_{l \in \mathcal{L}} a_m^l(t) \leq 1\}, \forall m \in \mathcal{M} \quad (8)$$

where $\mathcal{K}$ and $\mathcal{L}$ denote all sub-channels and users, respectively.

The network operates in discrete time slots. Communication parameters are constant per slot; each UAV holds CSI of all users for a decision cycle $T_s \geq 1$, makes resource allocation decisions, and completes transmission within t to $t + T_s$ before updating the next strategy.

2.3 Multi-UAV Network Model

We first clarify the optimization problem and model the joint user assignment, power level selection, and subchannel allocation of UAVs as a Markov Decision Process (MDP) to capture environmental uncertainty. The channel and signal models inform the communication environment and reward calculation for resource allocation efficiency.

In the network, $\mathcal{M}$ UAVs (each as an independent agent) operate without prior environmental information, non-cooperatively selecting actions at each time slot t to maximize long-term rewards. The action space of UAV m is $\Theta_m = \mathcal{A}_m \otimes \mathcal{C}_m \otimes \mathcal{P}_m$ (Cartesian product of user set $\mathcal{A}_m$, sub-channel set $\mathcal{C}_m$, and power level set $\mathcal{P}_m$), with action $\theta_m(t) = (a_m(t), c_m(t), p_m(t)) \in \Theta_m$ (selected user, sub-channel, power level). Let $\theta_{-m}(t) \in \Theta \setminus \Theta_m$ denote the action space of other UAVs.

To ensure communication reliability, the core objective is to satisfy a predefined QoS threshold for UAV SINR:

$$\gamma_m(t) \geq \bar{\gamma}, \forall m \in \mathcal{M} \quad (9)$$

where $\bar{\gamma}$ is the QoS threshold. The reward function $r_m(t)$ for UAV m is defined as throughput minus power consumption cost (zero if the QoS constraint is violated):

$$r_m(t) = \begin{cases} \frac{W}{K} \log(1 + \gamma_m(t)) - \omega_m P_m(t), & \text{if } \gamma_m(t) \geq \bar{\gamma}, \\ 0, & \text{o.w.} \end{cases} \quad (10)$$

Here, ω_m is the unit power cost, W is total bandwidth, K is the number of sub-channels (each occupying $\frac{W}{K}$ bandwidth), and $\gamma_m(t)$ is UAV m 's total SINR (defined in 4). The reward depends on UAV m 's selected action and channel gain $G_{m,l}^k(t)$, ignoring fixed power consumption (e.g., control/data processing).

UAV m aims to maximize the long-term discounted reward:

$$v_m(t) = \sum_{\tau=0}^{+\infty} \delta^\tau r_m(t + \tau) \quad (11)$$

where $\delta \in (0, 1]$ is the time discount factor (balancing short/long-term gains). The optimization problem for UAV m is to find the optimal action $\theta_m^*(t)$:

$$\theta_m^*(t) = \arg \max_{\theta_m \in \Theta_m} v_m(t) \quad (12)$$

The multi-UAV network optimization thus consists of M sub-problems, each corresponding to the strategy selection of one UAV.

3 System implementation

This section constructs a resource allocation method for UAV networks, named ASAC, based on the UAV network communication environment and UAV network model. Firstly, a MARL method is established based on UAV network communication, where each UAV operates an agent to make resource allocation decision, and the efficiency of model is improved through iterative training. Secondly, the adaptive spatial discount mechanism and the agent context-aware communication mechanism are added to the multi-agent reinforcement learning framework to enhance the convergence speed of the model and the efficiency of resource allocation.

3.1 Multi-Agent Reinforcement Learning Framework for UAV Communication Networks

Figure 2 illustrates the core components of the MARL framework for UAV communication networks, where each UAV m acts as an independent agent responsible for communication-related decision-making.

For UAV m , the state s_m^t includes its channel occupancy status (a K -dimensional 0-1 vector indicating sub-channel occupancy) and position. Agent m receives environmental reward r_m^t and executes action $\theta_m = (a_m, c_m, p_m)$ (user, sub-channel, power level selection). In multi-agent systems, each agent's Q-function is influenced by others' actions, making joint optimal strategy discovery challenging. We adopt an independent agent approach [16]: all UAVs share a deep neural network, learn independently based on local observations and rewards, and do not model other agents' behaviors, simplifying the learning process.

The resource allocation problem is solved via a MARL framework with Actor-Critic (A2C) networks. Each UAV learns the optimal

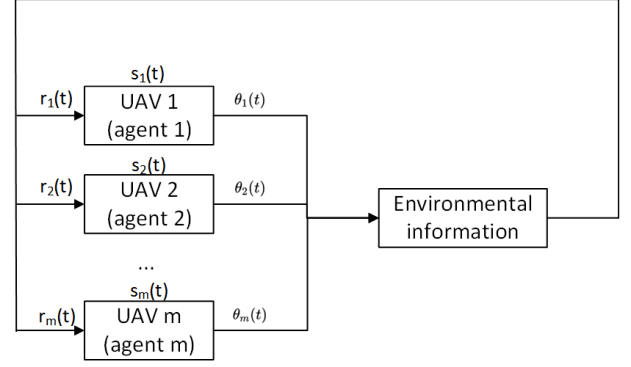


Figure 2: Framework of MARL for UAV communication networks

Q-value to determine its strategy, with the Q-value update rule:

$$Q_m^{t+1}(s_m, \theta_m) = Q_m^t(s_m, \theta_m) + \alpha \left\{ r_m^t + \delta \max_{\theta'_m \in \Theta_m} Q_m^t(s'_m, \theta'_m) - Q_m^t(s_m, \theta_m) \right\} \quad (13)$$

where $s_m = s_m^t$, $\theta_m = \theta_m^t$, $s'_m = s_m^{t+1}$, $\theta'_m = \theta_m^{t+1}$, and key parameters are defined in Table 1.

Table 1: Parameters in Q-value update rule

$r_m(t)$	Reward function of UAV m at time t
s_m^t	State of UAV m at time t
θ_m^t	Action of UAV m at time t

An ϵ -greedy strategy is adopted for exploration-exploitation:

$$\pi_m(s_m, \theta_m) = \begin{cases} 1 - \epsilon, & \text{if } Q_m \text{ of } \theta_m \text{ is the highest,} \\ \epsilon / (|\Theta_m| - 1), & \text{otherwise.} \end{cases} \quad (14)$$

The MARL algorithm for UAV networks is detailed in Algorithm 1, where T is the total time steps, and $\mathcal{M}$ is the set of UAVs. Each UAV initializes the DQN network and policy, then iteratively selects actions, calculates rewards based on SINR and QoS thresholds, updates Q-values and policies, until $t = T$.

Define N_m as the neighborhood of agent m , $H_m = N_m \cup \{m\}$ (local region), and $d(m, n)$ as the minimum edge distance between agents. To incorporate local interaction, the input state is extended to s_{m,H_m}^t (states of m and its neighbors). The local value function is:

$$V_m^t(s_{m,H_m}^t) = r_m^t + \delta V_m^{t+1}(s_{m,H_m}^{t+1}) \quad (15)$$

Due to communication delays, global information sharing is infeasible, and communication is limited to local regions. The value loss function (batch size $|B|$) is:

$$L(w_m) = \frac{1}{2|B|} \sum_{t \in B} \left(r_m^t + \delta V_m^{t+1}(s_{m,H_m}^t) - V_m^t(s_{m,H_m}^t) \right)^2 \quad (16)$$

The policy loss function (advantage function $A_m^t = Q_m^t - V_m^t$) is:

$$L(\theta_m) = -\frac{1}{|B|} \sum_{t \in B} \log \pi_{\theta_m}(\theta_m^t | s_{m,H_m}^t) A_m^t(s_{m,H_m}^t, \theta_m^t) \quad (17)$$

Algorithm 1 MARL Algorithm for UAV Networks

Input: $t = 0$, $\delta = 0.9$, QoS threshold $\bar{y}_m$

```

1: for  $m \in \mathcal{M}$  do
2:   Initialize  $Q_m^t(s_m, \theta_m)$ , policy  $\pi_m(s_m, \theta_m) = 1/|\Theta_m|$ 
3:   Initialize state  $s_m = s_m^t = 0$ 
4: end for
5: while  $t < T$  do
6:   for  $m \in \mathcal{M}$  do
7:     Select action  $\theta_m$  via  $\pi_m(s_m)$ 
8:     Calculate SINR  $y_m(t)$  using 2
9:     if  $y_m(t) \geq \bar{y}_m$  then
10:       $r_m(t) = \frac{W}{K} \log(1 + y_m(t)) - \omega_m P_m(t)$  (10)
11:    else
12:       $r_m(t) = 0$  (10)
13:    end if
14:    Update  $Q_m^{t+1}(s_m, \theta_m)$  via 13
15:    Update policy  $\pi_m(s_m, \theta_m)$  via 14
16:     $t = t + 1$ ,  $s_m = s_m^t$ 
17:   end for
18: end while

```

3.2 Multi-Agent Actor-Critic Framework Based on Adaptive Spatial Reward Mechanism

Through a series of local state transition processes, the impact of each agent on the global state is transmitted layer by layer in the system. To enhance the stability of the MARL algorithm and its adaptability to the environment, this study introduces an adaptive spatial reward mechanism, which dynamically adjusts the influence factor that each agent receives from other agents based on the distance between different agents, reducing the impact of distant reward signals on the agents, thereby making the model more compact and easier to converge.

The method integrates information from neighbor policies to enhance the visibility of each local agent. Additionally, a spatial discount factor is introduced to decrease the impact of states and reward signals from other agents on the agent in the question. In the input of the agent, in addition to the current state s_m , the latest policy sampled from the neighbor policies is also included. The calculation of this policy sampling is shown as follows:

$$\pi_m^{t+1} = \pi_{\theta_m}(\theta_m^t | s_{m,H_m}^t, \pi_{N_m}^t) \quad (18)$$

Where s_{m,H_m}^t represents the state of agent m and its neighboring area agents H_m , and $\pi_{N_m}^t$ represents the policy of neighborhood agents of agent m N_m (excluding agent m itself). The method focuses on utilizing the current time rather than long-term neighbor strategies. This is because the network state changes slowly in the short term, so the policy at the current time is highly correlated with the policy at the previous time, making the policy at the current time approximate the policies of many adjacent moments.

Although each agent has knowledge of the local area's state and neighborhood policy, fitting the global state remains challenging. To promote global cooperation, the global reward can be aggregated as $\bar{r}_m = \sum_{i \in H_m} r_i$. Based on this, a spatial discount factor κ is introduced to adjust the global reward of agent m to:

$$\bar{r}_m = \sum_{d=0}^{D_m} \left(\sum_{j \in H_m | d(i,j)=d} \kappa^d r_m \right) \quad (19)$$

Where D_m is the maximum distance that agent m can sense surrounding agents. κ is the spatial discount factor, which by reducing the influence of signals from agents that are farther away, gives more priority to the reward values of nearby agents, thus enhancing the stability of the model and accelerating its convergence. Compared to sharing the same global reward among all agents, the adaptive spatial reward mechanism can achieve a more flexible balance between greedy control ($\kappa=0$) and cooperative control ($\kappa=1$). When κ approaches 0, each agent tends to act individually, executing local greedy control; when κ approaches 1, agents tend to coordinate with each other, implementing a global cooperative strategy.

The global value is $\bar{V}_m^t = \sum \tau = t^\infty \delta^{\tau-t} \bar{r}_m^\tau$, which is the sum of the discounted values of all future global rewards for agent m . The calculation of the global value function is the sum of the current global reward and the discounted next-step global value, and its expression is:

$$\bar{V}_m^t(s_{m,H_m}^t, \pi_{N_m}^t) = \bar{r}_m^t + \delta \bar{V}_m^{t+1}(s_{m,H_m}^{t+1}, \pi_{N_m}^{t+1}) \quad (20)$$

Similarly, the expression for the global Q-function is:

$$\bar{Q}_m^{t+1}(s_{m,H_m}, \pi_{N_m}, \theta_m) = \bar{Q}_m^t(s_{m,H_m}, \pi_{N_m}, \theta_m) + \alpha \{ r_m + \delta \max_{\theta'_m \in \Theta_m} \bar{Q}_m^t(s'_{m,H_m}, \pi'_{N_m}, \theta'_m) - \bar{Q}_m^t(s_{m,H_m}, \pi_{N_m}, \theta_m) \} \quad (21)$$

The global value loss function for agent m is the residual of the current global reward value of agent m plus the global value of the current action, and its expression is:

$$\bar{L}(w_m) = \frac{1}{2|B|} \sum_{t \in B} \left(\bar{r}_m^t + \delta \bar{V}_m^t(s_{m,H_m}^t, \pi_{N_m}^t) - \bar{V}_m^{t+1}(s_{m,H_m}^{t+1}, \pi_{N_m}^{t+1}) \right)^2 \quad (22)$$

This update method makes the global value function more relevant to the local area observations ($s_{m,H_m}^t, \pi_{N_m}^t$) by inputting the other agents' strategies $\pi_{N_m}^t$ into the value function $\bar{V}_m^t$, and the calculation of the global reward $\bar{r}_m^t$ uses a spatial discount factor, thereby making the model more stable.

The global policy loss function is expressed as:

$$\begin{aligned} \bar{L}(\theta_m) = & - \frac{1}{|B|} \sum_{t \in B} \log \pi_{\theta_m}(\theta_m^t | s_{m,H_m}^t, \pi_{N_m}^t) \bar{A}_m^t(s_{m,H_m}^t, \pi_{N_m}^t) \\ & - \beta \sum_{n \in H_m} \pi_{\theta_n} \log \pi_{\theta_n}(\theta_n^t | s_{n,H_n}^t, \pi_{N_n}^t) \end{aligned} \quad (23)$$

Where $\bar{A}_m^t(s_m, \theta_m)$ is the global advantage function, which is the difference between the global Q function and the global value function, expressed as $\bar{A}_m^t(s_m, \theta_m) = \bar{Q}_m^t(s_{m,H_m}^t, \pi_{N_m}^t, \theta_m^t) - \bar{V}_m^t(s_{m,H_m}^t, \pi_{N_m}^t)$. The additional regularization term introduces an extra entropy function into the policy $\pi_{N_m}^t$ to encourage early exploration.

Algorithm 2 shows the MA2C algorithm under the synchronous update condition. κ represents spatial discount factor, δ denotes time discount factor, T represents the time range of each iteration, $|B|$ represents the size of the mini-batch, and η_w represents the learning rates for the critic, η_θ represents the learning rates of the actor networks. In the initial phase, each local agent follows the current strategy to collect experiences until enough samples are accumulated for mini-batch updating, as detailed in lines 5-10. If the mini-batch sample collection process is terminated early in this iteration, the agent will restart a completely new round, as shown in lines 11-13. It is necessary to note that early termination may affect the estimation of reward values. If the trajectory reaches the termination state at time T exactly, the reward value (seen in line 16) needs to be adjusted to:

$$\bar{V}_{t,m} = \begin{cases} \sum_{\tau=t}^{T-1} \delta^{\tau-t} \bar{r}_{\tau,m} & \text{before reset,} \\ \sum_{\tau=t}^{\infty} \delta^{\tau-t} \bar{r}_{\tau,m} & \text{after reset.} \end{cases} \quad (24)$$

Next, the critic and actor networks are updated using mini-batch gradients, as shown in lines 17 and 18. Common gradient optimizers uses Stochastic Gradient Descent (SGD). During training, if the predefined max step size is arrived or the termination condition is satisfied, the training process will be terminated.

Algorithm 2 Multi-agent A2C based on spacial discount algorithm

Input:

- 1: parameters: $\alpha, \delta, \kappa, |B|, T, \eta_w, \eta_\theta$.
- 2: initialize $s_0, k = 0, t = 0, B = \emptyset$
- 3: output: $\{w_m\}_{m \in \mathcal{V}}, \{\theta_m\}_{m \in \mathcal{V}}$

Output:

```

4: repeat
5:   for m in v do do
6:     sample  $\theta_{t,m}$  from  $\pi_{t,m}$ ;
7:     receive  $r_{t,m}$  and  $s_{t,m}$ ;
8:   end for
9:    $B = B \cup (t, s_{t,m}, \pi_{t,m}, u_{t,m}, r_{t,m}, s_{t+1,m})_{m \in \mathcal{V}}$ ;
10:  t=t+1, k=k+1;
11:  if t=T then
12:    initialize  $s_0, t = 0$ ;
13:  end if
14:  if k=|B| then
15:    for m in  $\mathcal{V}$  do
16:      estimate  $V_{\tau,m}, \forall \tau \in B$ ;
17:      update  $w_m$  with  $\eta_w \nabla L(w_m)$ 
18:      update  $\theta_m$  with  $\eta_\theta \nabla L(\theta_m)$ 
19:    end for
20:    k = 0, B =  $\emptyset$ 
21:  end if
22: until reach stop condition

```

3.3 Multi-Agent Actor-Critic Framework Based on Agent Context-aware Communication Mechanism

To address slow convergence and poor performance of traditional non-communicative MARL algorithms (e.g., IA2C, FPrint, ConSeNet), we propose an agent context-aware communication mechanism: agents exchange encoded state information with neighbors to improve learning efficiency.

3.3.1 Communication Mechanism. Define the observation space as $o_{m,t} = s_m^t \cup g_m^t$, where s_m^t is agent m 's local state, and g_m^t is communication messages from neighbors $n \in N_m$.

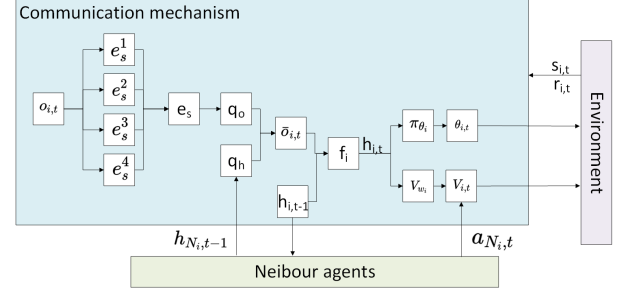


Figure 3: Communication mechanism

Figure 3 illustrates the mechanism: input $o_{m,t}$ is processed via differentiable encoding/extraction functions (e_s, q_o, q_h) and LSTM encoder f_l to generate hidden state $h_{m,t}$, which updates policy π_{θ_m} and value function V_{w_m} . The hidden state update rule is:

$$h_{m,t} = f_l(h_{m,t-1}, q_o(e_s(o_{m,t})), q_h(h_{N_m,t-1})) \quad (25)$$

where $h_{m,t-1}$ is agent m 's previous hidden state, $h_{N_m,t-1}$ is neighbors' previous hidden states (i.e., g_m^t). e_s encodes $o_{m,t}$ (regrouped by physical units and independently encoded before concatenation), q_o/q_h extract features, and f_l (64-neuron LSTM) fuses temporal and communication information.

The concatenated representation $\delta_m^t = \text{cat}(q_o(e_s(o_{m,t})), q_h(h_{N_m,t-1}))$ reduces information loss, and $h_{m,t}$ drives stochastic action sampling ($a_m^t \sim \pi_{\theta_m}(\cdot | h_m^t)$) and value prediction. Neighbor action information is further introduced into the critic network (inspired by MADDPG) to enhance training.

3.3.2 Action Smoothing. To reduce action randomness and stabilize training, we apply Exponential Moving Average-based action smoothing:

$$a_m^t \leftarrow \begin{cases} a_m^t, & t = 1 \\ \rho a_m^{t-1} + (1 - \rho) a_m^t, & t > 1 \end{cases} \quad (26)$$

where $\rho \in [0, 1]$ is a discount factor (hyperparameter tuned via cross-validation). A smoothing window T_w stores historical actions—excessively large T_w slows responsiveness, while small T_w fails to smooth effectively.

The complete algorithm is depicted in Algorithm 3. Hyperparameters are as follows: Action smoothing factor ρ , time discount factor γ , learning rates for the actor network η_w and the critic network η_θ , total number of training steps M , training batch size T , action sampling window T_w , and batch size N . Agents interact with the environment in thousands of batches, as seen in lines 4-31. Each agent gathers communication messages $g_{m,t}$ and sends them to its neighbors, as seen in lines 6-8. Following this, each agent combines, regroups and encodes its observations, as seen in line 10. Then, the agents update their hidden states and current policies, after which they sample actions from the probability distribution corresponding to the policy, as seen in lines 11-12. Subsequently, state values are updated, and before execution, actions are smoothed, as

seen in lines 14-17. After interacting with the environment (seen in line 17), agents are transmitted to the next state, then receive immediate rewards (seen in line 18), which are added to the experience buffer, as seen in line 19. At the end of each episode, actor and critic network parameters are updated using sampled data from the experience buffer, as seen in lines 21- 27. If the episode is complete, the code marks DONE. Upon receiving DONE, all agents reset and then begin another batch, as seen in line 29.

Algorithm 3 Multi-agent A2C based on communication mechanism algorithm

Input: $\alpha, \delta, \kappa, |B|, T, \eta_w, \eta_\theta$
Output: θ_m, w_m for $m \in \mathcal{M}$

```

1: Initialize  $s_0, t = 0$ , buffer  $\leftarrow \emptyset$ 
2: for  $j = 0$  to  $N$  do
3:   for  $t = 0$  to  $T$  do
4:     for  $i \in \mathcal{M}$  do
5:       Message  $g_{i,t} = h_{i,t-1}$ 
6:     end for
7:     for  $m \in \mathcal{M}$  do
8:        $\tilde{o}_m^t = \text{cat} (q_o (e_s (o_{m,t})), q_h (h_{Nm,t-1}))$ 
9:        $h_{m,t} = f_m (h_{m,t-1}, q_o (e_s (o_{m,t})), q_h (h_{Nm,t-1}))$ 
10:       $\pi_{m,t} = \pi_{\theta_m} (\cdot | h_{m,t}); a_{m,t} \sim \pi_{m,t}$ 
11:    end for
12:    for  $m \in \mathcal{M}$  do
13:      Update  $a_{m,t}$  via Eq26; execute  $a_{m,t}$ 
14:    end for
15:    Simulate  $\{s_{m,t+1}, r_{m,t}\}_{m \in \mathcal{M}}$ 
16:    buffer  $\leftarrow \{(o_{m,t}, a_{m,t}, r_{m,t}, v_{m,t})\}_{m \in \mathcal{M}}$ 
17:     $t = t + 1, j = j + 1$ 
18:  if Done then
19:    for  $m \in \mathcal{M}$  do
20:      Update  $\theta_m \leftarrow \theta_m + \eta_{\theta_m} \nabla_{\theta_m} J (\theta_m)$ 
21:      Update  $w_m \leftarrow w_m + \eta_{w_m} \nabla_{w_m} V (w_m)$ 
22:    end for
23:    buffer  $\leftarrow \emptyset$ 
24:  end if
25: end for
26: Update  $s_0, t = 0$ 
27: end for
```

4 Experiment

4.1 Environmental Settings

Simulation tests the resource allocation efficiency of the proposed method. We consider a multi-UAV network deployed in a 500m-radius disk area, where 200 ground users are uniformly randomly distributed in a ring. All UAVs fly at a fixed altitude $H = 100$ m, with 6 UAVs forming an aerial network to provide communication services. Signal transmission adheres to the established UAV-to-ground channel and signal models. For UAV m at time slot t , state s_m denotes sub-channel occupancy, and action $\theta_m(t) = (a_m(t), c_m(t), p_m(t))$ represents selected user, sub-channel, and power level. The reward r_m reflects communication quality, and the goal is to maximize the total long-term reward (i.e., resource allocation efficiency).

Simulation parameters are set as follows: sub-channel bandwidth $\frac{W}{K} = 75$ kHz (W = total bandwidth, K = number of channels); transmission completion threshold $T_s = 0.1$ s. For the probabilistic channel model (referring to [2]), LoS probability parameters $a = 9.61$, $b = 0.16$; carrier frequency $f = 2.3$ GHz; additional path losses $\eta^{\text{LoS}} = 1$, $\eta^{\text{NLoS}} = 20$. For the LoS model: reference distance (1m) channel gain $\beta_0 = -60$ dB, path loss exponent $\alpha = 2$. Other parameters: maximum power levels $J = 4$, UAV maximum power $P_m = 23$ dBm (equally divided into J discrete values); unit power cost $\omega_m = 100$; minimum user SINR threshold $\gamma_0 = 3$ dB. Detailed parameters are listed in Table 2.

Table 2: Simulation parameters

Parameter	Value
Sub-channel bandwidth $\frac{W}{K}$	75 KHz
Los connection parameter a	9.6
Los connection parameter b	0.15
Carrier frequency f	2.3 GHz
LoS channel gain parameter η^{LoS}	1.2
NLoS channel gain parameter η^{NLoS}	22
Reference distance d_m	1 m
Reference distance channel power gain β_0	-60 dB
Path loss coefficient α	2
Maximum power level number J	4
Maximum power of each UAV $W_n = W$	23 dBm
Cost per power level $\omega_n = \omega$	120
Minimum user SINR γ_0	2.7 dB

At the start of each training round, the user positions are randomly initialized, and the UAV positions are initialized using the K-means method, i.e., the UAV positions are initialized over the user clustering areas, forming the initial environment for each training round. Thereafter, in each training round, each UAV selects an action based on the current state and receives the corresponding reward and the next state, performing a total of 400 steps of actions, after which the current round of training ends.

4.2 Hyperparameters setting

This subsection analyzes the impact of hyperparameter ϵ (exploration factor) on model performance to identify its optimal value.

Model parameters are fixed as: $\delta = 0.99$, actor learning rate $\eta_\theta = 5 \times 10^{-4}$, critic learning rate $\eta_w = 2.5 \times 10^{-4}$. Fair comparison is ensured via consistent random seeds per round, time slot $t = 0.05$ s (one decision per slot), and epoch duration $T = 20$ s.

Figures 4 and 5 plot average rewards for different ϵ values (UAV speed = 40 m/s) under LoS and probabilistic channel models, respectively (settings: $\mathcal{M} = 4$ UAVs, $K = 3$ subchannels, $\mathcal{L} = 200$ users). Both figures show similar reward trends: average reward rises then falls as ϵ increases from 0 to 1 (extreme ϵ harms training). The model achieves optimal average reward at $\epsilon = 0.5$ (balanced exploration).

4.3 Compared to State-of-the-art Benchmarks

The ASAC method proposed in this article is compared with several mainstream benchmark MARL algorithms (IA2C, MAB [20],

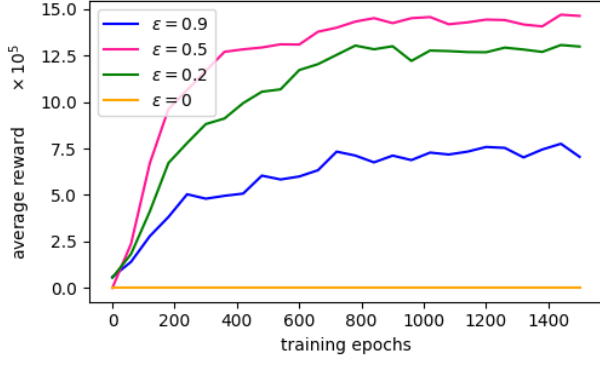


Figure 4: Average rewards of LoS channel of different ϵ for UAV network ($M=2, K=3, L=100$)

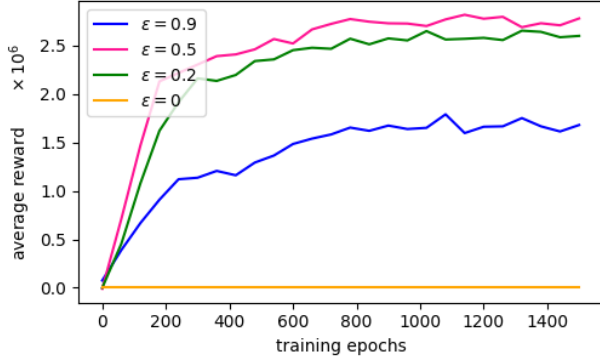


Figure 5: Average rewards of possibility model channel of different ϵ for UAV network ($M=5, K=4, L=250$)

MADDPG[14]), and the results demonstrate the effectiveness of the proposed method. Figure 6 shows the training curves for $M=4, K=3, L=50$ (small number users network) and $M=7, K=5, L=200$ (large number users network), respectively. In the experiment, the total reward represents resource allocation efficiency. To ensure a fair comparison, the baseline methods and the proposed method use the same time discount factor, actor network learning rate and critic network learning rate. Moreover, different random seeds are generated for each round, and all algorithms use the same random seeds for each round to ensure they have the same training and testing environments. The time slot t and the duration of each round T follow the settings mentioned above.

The proposed method ASAC outperforms all the benchmark MARL algorithms in terms of performance and convergence speed. Compared to MADDPG, the method improves communication efficiency and accelerates learning speed by introducing an adaptive spatial reward mechanism and an agent context-awareness mechanism. Particularly in complex scenarios with more users and more UAVs, the method further highlights its significant advantages in high number users scenarios due to its faster convergence speed and higher average reward (as shown in Figure 6b). At the same time, compared to the communication mechanism of MAB, the communication mechanism of this method is more efficient. Compared to

the optimal MAB among the baseline methods, this method slightly improves communication efficiency in small-scale networks; in high number users networks, the best baseline method MAB reach the total reward of 4.23, while this method reach 4.89, which gets a promotion of 16%.

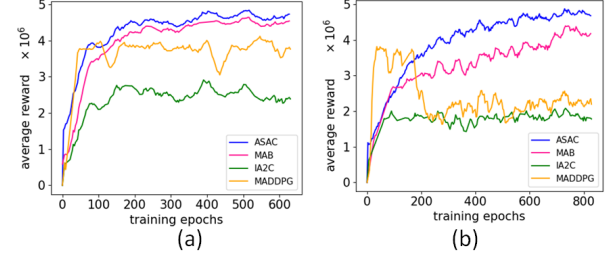


Figure 6: Average reward values of network of different scale. (a) small number users network. (b) large number users network.

Figure 7 displays the network communication efficiency under different numbers of users. Through simulations, the average network utility of each method across 100 test rounds with various user numbers is obtained. This method consistently outperforms the benchmark methods in aspects of network utility across all user numbers. When the number of users enhances, the total capacity of the UAV network also grows. Concurrently, to assure coverage of the network, UAVs need to increase their transmission power. Within the network communication capacity, this method adjusts the resource allocation strategy to mitigate the interference and energy consumption effects resulting from the increased power, thus maximizing network utility.

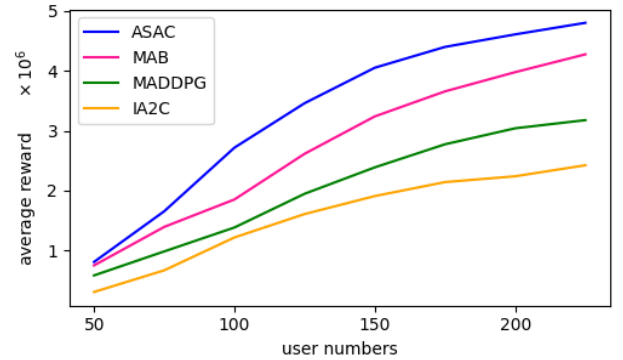


Figure 7: Average reward values of the UAV network under different numbers of users ($M=4, K=3$)

5 Conclusion

The article deeply investigates the issues of model convergence difficulties and low communication efficiency in multi-UAV communication networks. Firstly, the communication problem of UAV

networks is modeled, establishing channel models and signal models for UAV-to-ground communication. Secondly, the method of ASAC is proposed, which effectively addresses the spatial correlation in multi-UAV systems by introducing an adaptive spatial reward mechanism, allowing UAVs to allocate resources more reasonably, and by introducing a communication mechanism with agent context-awareness, adjacent UAVs can exchange state information and strategies in real-time, thereby promoting global cooperation, accelerating model convergence speed, and improving model performance. Finally, experimental results show that the method proposed in the article outperforms other baseline algorithms in terms of convergence speed and resource allocation efficiency, especially in high number users networks and in the situation with high number of users, where the method increases resource allocation efficiency by 16%.

Acknowledgments

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